

10	A	At steady state bulb X and Y have the same pressure, so $\frac{V_X}{n_X R T_X} = \frac{V_Y}{n_Y R T_Y}$ giving $\frac{n_X}{n_Y} = \frac{V_X}{V_Y} \frac{T_Y}{T_X}$, but for the same gas, $n \propto m$ (since $n = \frac{m}{m_R}$) Hence $\frac{m_X}{m_Y} = \frac{V_X}{V_Y} \frac{T_Y}{T_X} = 2 \left(\frac{400}{200} \right) = 4$ so, $m_Y = \frac{m}{4}$
11	B	work done = $n \Delta V = n (V_2 - V_1) = n V_2 - n V_1 = n R T_2 - n R T_1$